X-ChemIR: A Patent-Grounded Benchmark Suite for Cross-Lingual Chemistry Retrieval

Anonymous ACL submission

Abstract

Chemical knowledge is often searched across language boundaries: a user may ask a technical question in one language while the strongest supporting document is written in another. Retrieval systems should therefore be tested on their ability to retrieve cross-language evidence, not only easier same-language matches. Yet multilingual chemistry retrieval has few clean public evaluation resources, and aggregate recall can conceal deployment-relevant failures. We address this gap with two public multilingual chemistry QAC retrieval releases derived from Google Patents and EPO data, together with a generation pipeline that preserves source, language, and quality metadata. The releases provide controlled multilingual technical evidence, while an auxiliary ontology-based diagnostic tests whether models confuse chemically plausible hard negatives with gold documents. Across the benchmark suite, aggregate Recall@10 overstates deployment readiness: the best cross-language Recall@10 is about 0.55, a same-language advantage remains visible for every query language, and chemically confusable documents outrank all gold documents on 14–48% of alias-graph queries. The benchmark turns cross-language and chemistry-confusability failures into measurable model-selection criteria for high-trust technical retrieval. Data and code are available on Hugging Face (Google Patents), Hugging Face (EPO), and GitHub.

1 Introduction and Related Work

Technical chemistry search rarely respects a single language boundary: a chemist may query in English, German, French, or Spanish while relevant evidence resides in a multilingual patent publication, a regional filing, or a document whose language differs from the query. This is not only a translation problem: chemistry retrieval must also distinguish a target compound from chemically related but incorrect alternatives. A retriever for a high-trust workflow such as patent search or RAG therefore has to do two hard things at once: cross the language barrier and resist chemical confusability.

Dense retrieval with neural sentence and passage embeddings (Reimers and Gurevych, 2019; Karpukhin et al., 2020) is attractive in this setting because multilingual encoders promise a single retrieval index across languages (Feng et al., 2022; Chen et al., 2024). However, recent work on cross-lingual retrieval and multilingual RAG shows that retrieval behavior can depend strongly on language alignment, translation effects, and same-language preference (Goworek et al., 2025; Amiraz et al., 2025; Li et al., 2024; Liu et al., 2025; Ki et al., 2025). Aggregate Recall@k can therefore overstate readiness when strong performance on easier same-language matches masks failures to retrieve cross-language evidence. It also does not reveal whether chemically related but incorrect documents outrank the intended evidence.

Existing benchmark lines cover important parts of this space. General embedding and multilingual retrieval benchmarks such as MTEB, MMTEB, MIRACL, NeuCLIR, and CLIRMatrix enable systematic comparison of retrieval and embedding models across languages (Zhang et al., 2023; Lawrie et al., 2023; Muennighoff et al., 2023; Enevoldsen et al., 2025; Sun and Duh, 2020). Chemistry-specific resources such as ChemTEB, ChemLit-QA, ChemComp, ChemKG-MultiHopQA, and ChEmbed extend evaluation to chemical embeddings, RAG, and scientific reasoning (Kasmaee et al., 2024; Wellawatte et al., 2024; Khodadad et al., 2026; Astaraki et al., 2026; Kasmaee et al., 2025). What remains missing is their combination: multilingual chemistry retrieval data with related technical evidence across languages, explicit same-language and cross-language relevance judgments, and chemically plausible hard negatives. Chemical ontology and alias resources such as ChEBI (Hastings et al., 2016) provide the structure needed to define such hard negatives. This use is complementary to ontology-grounded extraction work such as CEAR (Langer et al., 2024), which extracts chemical entities and roles rather than testing retrieval against chemically related look-alikes.

Patents are a practical substrate for this evaluation. Patent retrieval has a long evaluation history, including NTCIR and CLEF-IP (Fujii and Ishikawa, 2002; Piroi et al., 2011), and recent work has revisited patent representation learning and embedding evaluation (Ghosh et al., 2024; Yousefiramandi and Cooney, 2026). Because patent publications often contain related technical disclosures across languages, they provide comparable multilingual technical evidence beyond prior-art search alone.

These gaps motivate X-ChemIR, a patent-grounded benchmark suite for cross-lingual chemistry retrieval. Our contributions are: (1) two public QAC retrieval releases derived from Google Patents and EPO data, with query/document-language metadata, plus an auxiliary ontology-based chemical-confusability stress test; (2) a reproducible LLM-assisted QAC generation and human-audit pipeline; (3) language-aware robustness diagnostics reported next to recall, including same- versus cross-language relevance, coverage depth, ranking consistency, reading cost, reranker recoverability, and alignment-only failures; and (4) an evaluation of eight off-the-shelf multilingual embedding models that quantifies cross-lingual degradation and chemical-confusion failures.

2 QAC Generation Pipeline

Our pipeline turns multilingual chemistry patent text into question–answer–context (QAC) records for retrieval evaluation. It produces questions in five languages, designed for cross-lingual retrieval and kept close to what a practitioner would actually search, in both technical and general modes. Figure 1 summarizes the generation and evaluation flow.

Corpus and context construction. We build on two patent-derived corpora that contribute complementary kinds of multilinguality. *Google Patents* supplies scale and language breadth: 23,787 published applications, predominantly WIPO (WO, 15,048) and European (EP, 4,610) filings, with a sizeable Mexican (MX, 3,952) share, spanning five languages (English, French, Spanish, German, Chinese), from which we take the title and abstract as the evidence block. On this side the non-English versions are Google's *machine* translations, and because Chinese coverage was zero we machine-translated 400 documents to seed it using GPT 5.5. A subset of Machine-translated documents (10%) are verified by Chinese native annotators to confirm the quality of the translation. More details on this topic is presented in ?? (supplementary Materials). *EPO* supplies clean cross-lingual parallelism: 11,315 granted B1 specifications, each published as a *human*-translated English/German/French triple (≈3,770 per language) with professionally translated first claims, which we add to the evidence block; we introduce no translations of our own on this side.

Candidate generation. For each source document we prompt `gpt-5-mini` to generate three candidate questions, each grounded in a completely by the passage. Table S.1 provide an overview of the dataset and its statistics.

Two choices shape the query distribution. First, the query *language*: the queries are generated in three set ups: (i) the question's *target language* is sampled per document relative to the languages that document exists in: *random-existing* picks a language the patent already has a version in, so the query has both a cross-lingual gold context and a same-language gold context; (ii) *random-missing* picks one it does *not*, yielding a pure cross-lingual "no-home" query, where no same-language gold-context is available for the query; and (iii) *all* generates the question in every 5 languages we targeted independent of availability of the gold-context document in those languages. Second, each question is written in one of two *modes*: *technical* questions ask for (i) a single concrete fact, (ii) a parameter, (iii) a material, (iv) an outcome, (v) method, or (vi) structural detail, answerable by the document, while *general* questions ask about the document's problem, approach, or application in deliberately paraphrased, low-overlap wording. The mode prompts are written in the target language to bring the generated questions closer to naitive quality. The prompts (English version) are available in Figures S.3 and S.4.

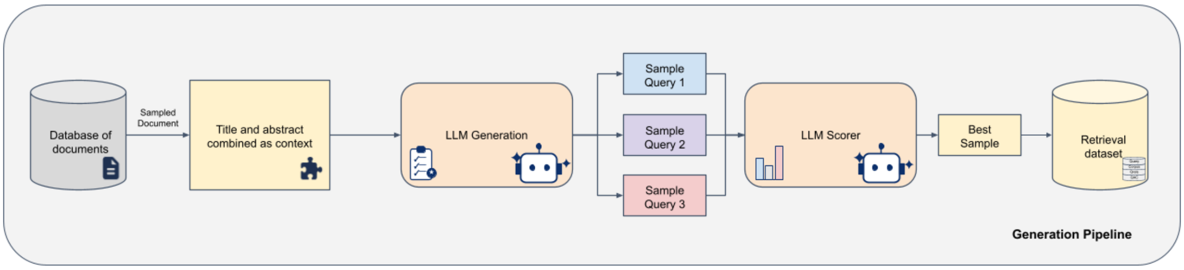

Figure 1: Overview of the QAC generation pipeline: source documents are converted into grounded candidate questions, scored, and exported as retrieval-ready records. Icons adapted from Font Awesome, licensed under CC BY 4.0.

Scoring, human audit. The three generated candidates per document for each query are scored not by the generator but by a separate verifier, `claude-sonnet-4.6`, which rates each candidate on multiple metrics around faithfulness and mode-specific quality; we keep the highest-scoring candidate per document as the selected query (more details on verifier prompts are available in Figures S.6–S.8). We then export the standard retrieval artifacts. In the process of query generation, the full generation pipeline benefited from three rounds of domain-expert annotation and feedback collection which significantly refined both the generation quality and the verifier prompts. In the final set of generated queries, the verifier was audited against human-annotatation on a 97-sample subset of the queries using criteria that mirror the LLM rubric. In this evaluation, the human feedback's mean score was **8.33/10** (technical: **8.72**, general: **7.91**) and *no* query was rejected (none scored in the 0–3 band; see Figure S.5 for more details). Also the agreement between LLM-verifier and human expert annotator was ?.

3 Evaluation Setup

We evaluate multilingual embedding models as drop-in dense retrievers, and design the setup to surface cross-lingual failure rather than average it away. Each query is judged against its source patent in *both* its own language and every other, separating same-language from cross-language relevance; an auxiliary alias-graph stress test adds chemically confusable hard negatives. We report a metric family that places capability, cross-linguality, and deployment cost side by side, and run it over a fixed set of eight models against the shared multilingual corpus.

Same- vs. cross-language relevance. A query is relevant to its source patent in every language that patent exists in, so we split each judgment into same-language relevance and cross-language relevance (a foreign-language version to the query). Some queries have no same-language gold at all, a built-in "no-home" cross-lingual stress test. We retain query language, document language, source office, and question mode (technical or general) as metadata, so every result can be sliced along each axis.

Alias-graph stress test. The alias graph asks a sharper question: given a concept named in several languages, can a retriever find the right document across languages while ignoring documents about chemically similar concepts? We build it from the ChEBI ontology (Hastings et al., 2016). For each compound we take a source patent passage that mentions it and prompt the generator with a technical-style prompt, but with the compound supplied, to write a question grounded in that passage whose answer is the compound itself, never naming the compound or any of its multilingual aliases (150 questions generated.). The gold is then the source publication and its parallel language versions, so a query in one language must reach that publication in another; the hard negatives are patents that mention a taxonomic neighbour of the compound (a sibling or parent in the graph) but not the compound itself: chemically close, but the wrong answer. These near-miss distractors operationalize, at the retrieval level, the distractor-latch failure that is destructive downstream, where a generator locks onto a plausible but wrong neighbour rather than the intended evidence (Astaraki et al., 2026).

Metrics. Alongside Recall@10 and its same- and cross-language split (MoLIR@10 and CLIR@10), we report measures aimed at cross-linguality. $\mathrm{LT\%} = \mathrm{CLIR@10}/\mathrm{MoLIR@10}$ is the share of a model's same-language ceiling it keeps cross-lingually (low LT signals language-siloing). For cross-lingual coverage we treat language as a retrieval facet: restricting to queries that have both gold types, $k^{\star}_{80}$ is the smallest depth at which the same-language gold document and the first cross-language gold document are present for 80% of queries, with halves k_{same} and k_{cross} giving that depth for the same- and the cross-language gold separately (censored at $>$1000). On the alias-graph benchmark we add cross-lingual RBO (rank-biased overlap; Webber et al., 2010), a consistency measure: it compares the top-k lists a model returns for the same concept asked in each of the five languages ($1 =$ identical, $0 =$ disjoint), so a low RBO means the same information need retrieves different documents depending on the query language. Three instruments read off deployment cost: XRC, the reading-cost multiplier $\mathrm{XRC}_C = D^{\mathrm{cross}}_C / D^{\mathrm{same}}_C$ (median XRC50), the factor by which one must read deeper to reach a cross-language gold than a same-language copy; $\mathrm{RRC@}K = \Pr[\text{first cross-language gold rank} \leq K]$, the re-ranker recoverability ceiling, so $1 - \mathrm{RRC@}K$ is unrecoverable by any top-K re-ranker and its budget curve has a knee $K^{\star}$; and ARI, which splits the shortfall at depth K into a re-rankable part $\mathrm{RRC@}K$, a deeper-pool part $\mathrm{RRC@1000} - \mathrm{RRC@}K$, and an alignment-only floor $L_{\infty} = 1 - \mathrm{RRC@1000}$ that no re-ranker over the retrieved top-1000 can move, reported as $\mathrm{ARI@}K = L_{\infty}/(1 - \mathrm{RRC@}K)$. All recall-based figures are language-balanced (a mean over query languages).

Models and protocol. We evaluate 8 multilingual or domain-relevant embedding models under 1 billion parameters: `embeddinggemma`, `bge-m3` (Chen et al., 2024), `granite-278m`, `nomic-v2-moe`, `qwen3-0.6B`, `LaBSE` (Feng et al., 2022), `SapBERT` (Liu et al., 2021), `e5-large-instruct`. All models retrieve against the same multilingual chemistry corpus, and all queries are evaluated with the same relevance judgments.

4 Results

4.1 Model Rankings and the Cross-Lingual Gap

embeddinggemma leads on both offices and on every axis (Google Patents / EPO Recall@10 0.57 / 0.58, CLIR@10 0.54 / 0.52) and reaches both golds the fastest ($k^{\star}_{80}$=147 vs. 304 for the next model), with bge-m3 a consistent second; the ordering barely changes between the two offices, so it reflects the task ratherqwen3-0.6B and nomic-v2-moe are mid-pack, while granite-278m, LaBSE and SapBERT trail by 20–30 Recall@10 points—and one model is an instructive outlier: e5-large-instruct looks mid-pack on raw Recall@10 (0.21 / 0.29) but its CLIR@10 collapses to 0.09 / 0.12 and its LT to 12 / 19, failing on cross lingual task. Off-the-shelf multilingual embedders are therefore far from interchangeable on this task, and even the best absolute numbers are low for an industrial deployment.

Across the whole field, retrieving a same-language copy of the answer is far easier than retrieving its foreign twin and no model escapes the drop. The cross-lingual coverage depths make the gap concrete: for the models that retrieve effectively the same-language gold surfaces within a few dozen documents (k_{same} 36–66) while its cross-language twin takes several hundred (k_{cross} 121–321); for the weaker models both depths balloon and the cross-language gold is frequently never covered for 80% of queries within the retrieved top-1000. The same pattern holds language by language: pooled across models, same-language recall stays near 0.56–0.62 in every query language while cross-language recall sits around 0.34–0.38, a roughly 0.25 gap present in all five languages (Figure S.1); and the query$\times$document grid shows the loss lives off the diagonal for every model (Figure S.2): even the strongest retrievers fall to a fraction of their same-language cell on individual routes, embeddinggemma drops from 0.69 same-language (English) to 0.18 when an English query must reach a German document, bge-m3 from 0.62 (German) to 0.33 reaching a Spanish document, and nomic-v2-moe from 0.69 (French) to 0.12 reaching a Chinese one. Cross-linguality, not raw retrieval power, is what every model struggles with.

	Google Patents			EPO			Cross-lingual coverage		
Model	R@10	CLIR	LT%	R@10	CLIR	LT%	$k^{\star}_{80}$	k_{same}	k_{cross}
embeddinggemma	**0.57**	**0.54**	73	**0.58**	**0.52**	74	**147**	**36**	**121**
bge-m3	0.51	0.47	**74**	0.55	0.47	66	304	66	215
qwen3-0.6B	0.49	0.45	72	0.47	0.39	61	425	63	310
nomic-v2-moe	0.46	0.41	63	0.51	0.43	63	367	55	321
granite-278m	0.41	0.39	72	0.39	0.36	84	>1000	641	569
LaBSE	0.30	0.27	59	0.27	0.25	**85**	>1000	761	>1000
e5-large-instruct†	0.21	0.09	12	0.29	0.12	19	>1000	47	>1000
SapBERT	0.24	0.20	49	0.22	0.17	54	>1000	807	>1000

Table 1: Retrieval performance on both patent offices. Best per column in bold.

4.2 Alias-Graph: Inconsistency and Confusion

Asking the same compound in all five languages does not return the same patents. The best cross-lingual RBO any model reaches is only 0.39 (embeddinggemma), a ceiling no model beats (Figure 2a), and it drops below 0.15 for the weakest retrievers: two language variants of the same query share only a minority of their top-ranked documents, so the embedding space is far from language-agnostic.

The second failure is chemical confusion: a wrong but chemically similar compound out-ranks every gold patent on 14–48% of queries depending on the model (Figure 2b, ALL column). The rate is strongly language-dependent, in the direction the cross-lingual setting predicts. Averaged over models it is lowest for English queries (≈21%) and highest for Chinese (≈39%), with German, French, and Spanish in between (≈28–30%); the same chemical question is markedly more likely to surface a look-alike above the right document when it is asked in a lower-resource language than in English. The two failures compound: a model that is both inconsistent across languages and easily confused by near-neighbours is exactly the one a chemistry team must not deploy.

4.3 Reading Cost and the Re-Ranking Ceiling

XRC asks a concrete operational question: starting from the top of the ranked list, how much further must a user read to reach the foreign-language version of the answer than to reach a same-language copy? Panel (a) of Figure 3 reports the median, and it is never one: the cheapest model still reads about twice as deep (LaBSE, 1.9×) and the rest three to six times deeper (bge-m3 3.5×, embeddinggemma 5.0×, nomic-v2-moe 6.0×). A low reading cost is not a virtue on its own, though, the cheapest readers here are the weakest retrievers, which read shallow only because they surface so little, so XRC has to be read together with capability (CLIR@10), not minimized in isolation. We drop e5-large-instruct from all three panels because it fails the CLIR@10 $<$ 0.10 degeneracy gate: it almost never retrieves a foreign twin at all, so its reading-depth ratio explodes (median 161×) and its recoverability curve rests on mostly censored ranks, the instruments are undefined for a model that does not retrieve the thing whose depth they measure.

The other two panels ask what a downstream re-ranker could do about the buried cross-language documents. A re-ranker only reorders the candidate pool it is handed, so RRC@K (panel b) the fraction of cross-lingual queries whose first cross-language gold (the answer's document in another language) has appeared by rank K, is a hard ceiling on what re-ranking can recover: anything past depth K is invisible to a top-K re-ranker. Two things stand out. First, the recoverable documents are front-loaded: the knee $K^{\star}$, the point of sharpest diminishing returns on the curve, sits at 2–5 for the strong models, so a shallow top-5 pool already captures the bulk of the cheaply recoverable cross-language gold (embeddinggemma RRC@5 $=$ 0.53); a deeper pool keeps helping but with falling efficiency per decade (RRC@100 $=$ 0.78, RRC@1000 $=$ 0.93), while the weak LaBSE and SapBERT reach their knee only at depth 75–150. Second, even an exhaustive re-ranker over the full retrieved pool hits a wall: panel (c) splits each model's cross-lingual shortfall into what a cheap top-10 re-rank recovers, what a deeper pool up to top-1000 adds, and an alignment-only floor beyond the top-1000, cross-language gold that first-stage retrieval never surfaces and that no re-ranker over the retrieved candidates can reach. That floor

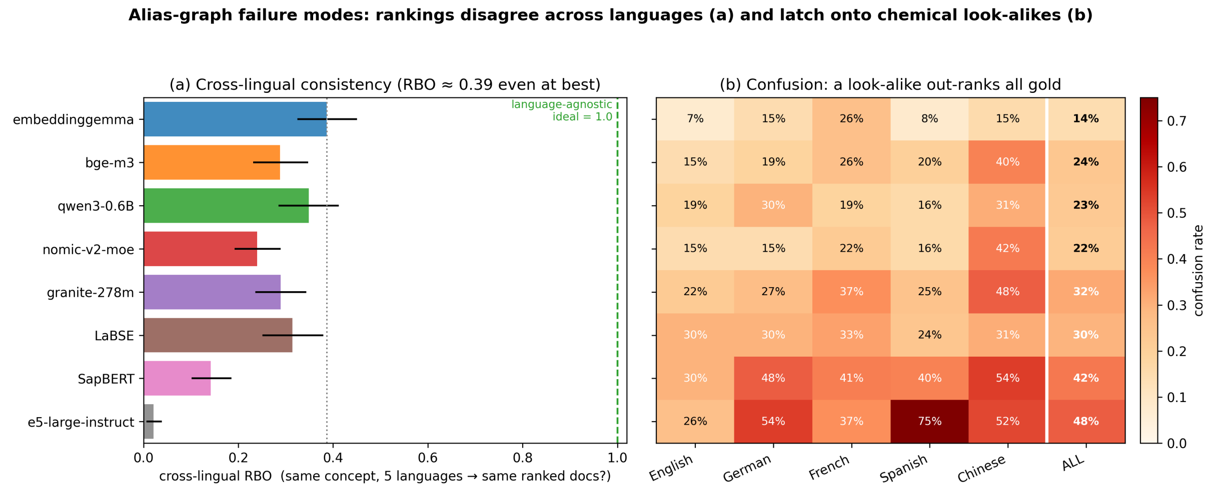

Figure 2: Alias-graph failure modes across the eight models. **(a)** Cross-lingual RBO: how much the top-ranked lists for the same compound asked in five languages overlap (1 = identical, 0 = disjoint); the most consistent model reaches only 0.39. **(b)** Confusion rate: how often a chemically confusable look-alike out-ranks every gold patent, by query language and pooled (ALL column); 14–48% overall, lowest for English and highest for Chinese.

is 7% of the gap for embeddinggemma and grows to 27% for SapBERT: the weaker the model, the more of its cross-lingual failure is structural. And a realistic top-100 re-ranker recovers even less, bounded by RRC@100 (0.78 for the best model), it reorders only what first-stage retrieval already placed in its pool, so the durable fix for the residual gap is representation alignment at indexing time, not a re-ranker at query time.

5 Conclusion

We introduced X-ChemIR, a patent-grounded benchmark suite for cross-lingual chemistry retrieval: two multilingual QAC benchmarks built from Google Patents and EPO patent data, an auxiliary alias-graph confusability stress test, and a robustness-metric family reported next to recall (same- versus cross-language relevance and LT, cross-lingual coverage depth $k^\star$, ranking consistency RBO, and the position-based metrics XRC, RRC, and ARI). Evaluating eight off-the-shelf multilingual embedders, we find that a single averaged recall hides the failure that ships: even the best model leaves nearly half of the cross-language evidence out of the top-10, a same-language home advantage persists in every query language, and a chemically confusable look-alike out-ranks the right document on 14–48% of alias-graph queries.

Limitations

The suite is intentionally specialized: chemistry patents give controlled multilingual technical evidence, but the results—and the model ordering—may not transfer to other domains, regulatory corpora, or general web retrieval, and patent variants, while content-controlled, are not guaranteed to be claim-level equivalent across every language, so the design is best read as a reusable evaluation pattern rather than a universal verdict. The QAC data rests on LLM generation and LLM-based verification; our 97-item human audit supports its quality but is limited and does not replace per-item expert chemistry or patent review, which we leave to future work (especially for borderline relevance and hard-negative severity). Finally, we report only the two QAC benchmarks and the alias-graph diagnostics; code-switching and noisy-query settings exist in the project but are not claimed here.

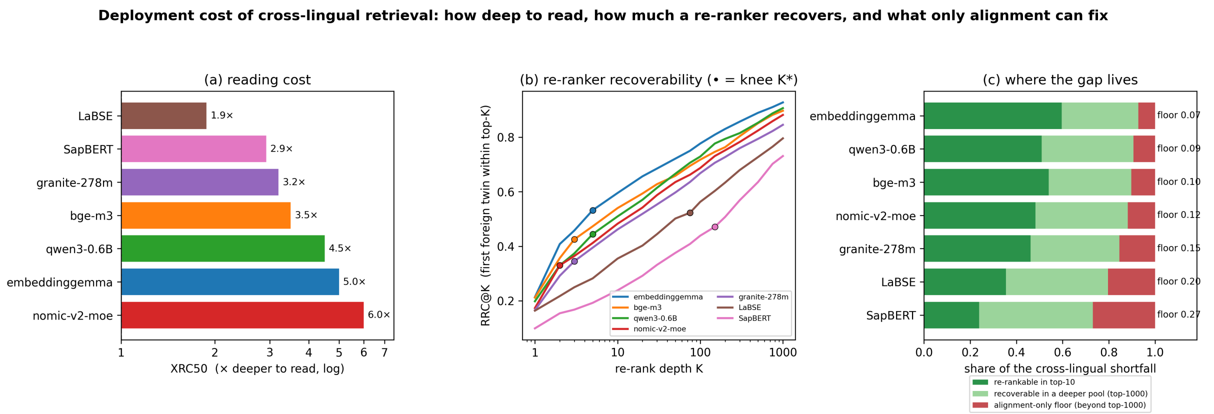

Figure 3: Deployment cost of cross-lingual retrieval (e5-large-instruct excluded by the CLIR@10 $<$ 0.10 gate). **(a)** XRC50 reading cost; **(b)** RRC@K re-ranker recoverability with knee $K^\star$ (•); **(c)** the shortfall split into re-rankable (top-10), deeper-pool (top-1000), and an alignment-only floor.

References

Chen Amiraz, Yaroslav Fyodorov, Elad Haramaty, Zohar Karnin, and Liane Lewin-Eytan. 2025. The cross-lingual cost: Retrieval biases in RAG over Arabic-English corpora. *arXiv preprint arXiv:2507.07543*.

Mahdi Astaraki, Mohammad Arshi Saloot, Ali Shiraee Kasmaee, Hamidreza Mahyar, and Soheila Samiee. 2026. When iterative RAG beats ideal evidence: A diagnostic study in scientific multi-hop question answering. *arXiv preprint arXiv:2601.19827*.

Jianlv Chen, Shitao Xiao, Peitian Zhang, Kun Luo, Defu Lian, and Zheng Liu. 2024. BGE M3-Embedding: Multi-lingual, multi-functionality, multi-granularity text embeddings through self-knowledge distillation. *arXiv preprint arXiv:2402.03216*.

Kenneth Enevoldsen and 1 others. 2025. MMTEB: Massive multilingual text embedding benchmark. *arXiv preprint arXiv:2502.13595*.

Fangxiaoyu Feng, Yinfei Yang, Daniel Cer, Naveen Arivazhagan, and Wei Wang. 2022. Language-agnostic BERT sentence embedding. *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (ACL)*, pages 878–891.

Atsushi Fujii and Tetsuya Ishikawa. 2002. NTCIR-3 patent retrieval experiments at ULIS. *arXiv preprint cs/0206035*.

Mainak Ghosh, Sebastian Erhardt, Michael E. Rose, Erik Buunk, and Dietmar Harhoff. 2024. PaECTER: Patent-level representation learning using citation-informed transformers. *arXiv preprint arXiv:2402.19411*.

Roksana Goworek, Olivia Macmillan-Scott, and Eda B. Özyiğit. 2025. What drives cross-lingual ranking? retrieval approaches with multilingual language models. *arXiv preprint arXiv:2511.19324*.

Janna Hastings, Gareth Owen, Adriano Dekker, Marcus Ennis, Namrata Kale, Venkatesh Muthukrishnan, Steve Turner, Neil Swainston, Pedro Mendes, and Christoph Steinbeck. 2016. ChEBI in 2016: Improved services and an expanding collection of metabolites. *Nucleic Acids Research*, 44(D1):D1214–D1219.

Vladimir Karpukhin, Barlas Oğuz, Sewon Min, Patrick Lewis, Ledell Wu, Sergey Edunov, Danqi Chen, and Wen tau Yih. 2020. Dense passage retrieval for open-domain question answering. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 6769–6781.

Ali Shiraee Kasmaee, Mohammad Khodadad, Mahdi Astaraki, Mohammad Arshi Saloot, Nicholas Sherck, Hamidreza Mahyar, and Soheila Samiee. 2025. ChEmbed: Enhancing chemical literature search through domain-specific text embeddings. *arXiv preprint arXiv:2508.01643*.

Ali Shiraee Kasmaee, Mohammad Khodadad, Mohammad Arshi Saloot, Nicholas Sherck, Stephen Dokas, Hamidreza Mahyar, and Soheila Samiee. 2024. ChemTEB: Chemical text embedding benchmark, an overview of embedding models performance and efficiency on a specific domain. In *Proceedings of The 4th NeurIPS Efficient Natural Language and Speech Processing Workshop*, volume 262 of *Proceedings of Machine Learning Research*, pages 512–531.

Mohammad Khodadad, Ali Shiraee Kasmaee, Mahdi Astaraki, Nicholas Sherck, Hamidreza Mahyar, and Soheila Samiee. 2026. Evaluating multi-hop reasoning in large language models: A chemistry-centric benchmark. In *Findings of the Association for Computational Linguistics: EACL 2026*, pages 6117–6143.

Dayeon Ki, Marine Carpuat, Paul McNamee, Daniel Khashabi, Eugene Yang, Dawn Lawrie, and Kevin Duh. 2025. Linguistic nepotism: Trading-off quality for language preference in multilingual RAG. *arXiv preprint arXiv:2509.13930*.

Stefan Langer, Fabian Neuhaus, and Andreas Nürnberger. 2024. CEAR: Automatic construction of a knowledge graph of chemical entities and roles from scientific literature. *arXiv preprint arXiv:2407.21708*.

Dawn Lawrie, Sean MacAvaney, James Mayfield, Paul McNamee, Douglas W. Oard, Luca Soldaini, and Eugene Yang. 2023. Overview of the TREC 2022 NeuCLIR track. In *Proceedings of the Thirty-First Text REtrieval Conference (TREC)*.

Bryan Li, Fiona Luo, Samar Haider, Adwait Agashe, Tammy Li, Runqi Liu, Muqing Miao, Shriya Ramakrishnan, Yuan Yuan, and Chris Callison-Burch. 2024. BordIRlines: A dataset for evaluating cross-lingual retrieval-augmented generation. *arXiv preprint arXiv:2410.01171*.

Fangyu Liu, Ehsan Shareghi, Zaiqiao Meng, Marco Basaldella, and Nigel Collier. 2021. Self-alignment pretraining for biomedical entity representations. In *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics (NAACL)*, pages 4228–4238.

Wei Liu, Sony Trenous, Leonardo F. R. Ribeiro, Bill Byrne, and Felix Hieber. 2025. XRAG: Cross-lingual retrieval-augmented generation. In *Findings of the Association for Computational Linguistics: EMNLP 2025*, pages 15669–15690.

Niklas Muennighoff, Nouamane Tazi, Loïc Magne, and Nils Reimers. 2023. MTEB: Massive text embedding benchmark. In *Proceedings of the 17th Conference of the European Chapter of the Association for Computational Linguistics (EACL)*, pages 2014–2037.

Florina Piroi, Mihai Lupu, Allan Hanbury, and Veronika Zenz. 2011. CLEF-IP 2011: Retrieval in the intellectual property domain. In *CLEF (Notebook Papers/Labs/Workshop)*. Multilingual (en/de/fr) patent prior-art retrieval; see also CLEF-IP 2009–2013 overviews.

Nils Reimers and Iryna Gurevych. 2019. Sentence-BERT: Sentence embeddings using Siamese BERT-networks. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pages 3982–3992.

Shuo Sun and Kevin Duh. 2020. CLIRMatrix: A massively large collection of bilingual and multilingual datasets for cross-lingual information retrieval. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 4160–4170.

William Webber, Alistair Moffat, and Justin Zobel. 2010. A similarity measure for indefinite rankings. *ACM Transactions on Information Systems*, 28(4):1–38.

Geemi P. Wellawatte, Huixuan Guo, Magdalena Lederbauer, Anna Borisova, Matthew Hart, Marta Brucka, and Philippe Schwaller. 2024. ChemLit-QA: A human evaluated dataset for chemistry RAG tasks. In *NeurIPS 2024 Workshop on Machine Learning and the Physical Sciences*.

Amirhossein Yousefiramandi and Ciaran Cooney. 2026. Benchmarking patent embeddings: A multi-task evaluation of 22 models across retrieval, classification, and clustering. *arXiv preprint arXiv:2605.24297*.

Xinyu Zhang, Nandan Thakur, Odunayo Ogundepo, Ehsan Kamalloo, David Alfonso-Hermelo, Xiaoguang Li, Qun Liu, Mehdi Rezagholizadeh, and Jimmy Lin. 2023. MIRACL: A multilingual retrieval dataset covering 18 diverse languages. *Transactions of the Association for Computational Linguistics*, 11:1114–1131.

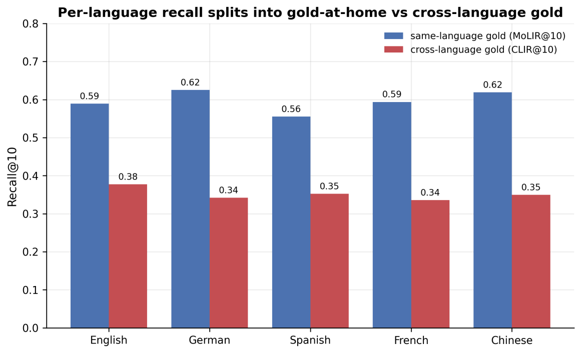

Figure S.1: Per query language, Recall@10 splits into same-language gold (MoLIR@10) and cross-language gold (CLIR@10), pooled over the non-degenerate models. The home advantage holds in every language.

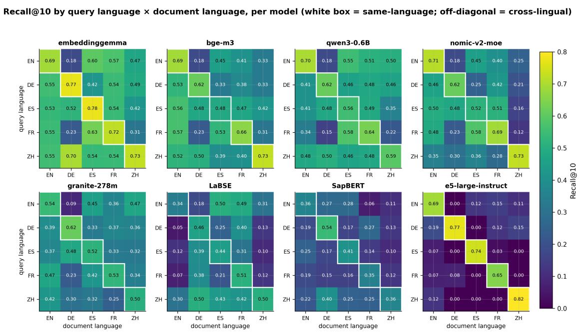

Figure S.2: Recall@10 by query language × document language, per model (white cell = same-language; off-diagonal = cross-lingual). The cross-lingual loss lives off the diagonal; e5-large-instruct's off-diagonal cells collapse to near zero.

A Appendix

This appendix collects supporting analyses that are useful for interpretation but too detailed for the six-page body.

A.1 Additional Cross-Lingual Retrieval Diagnostics

These diagnostics expand the paper's main cross-language retrieval result. They show that the same-language advantage is not a single-score artifact: it appears as query-language variation, same-language gold advantage, transfer across patent offices, and weaker score separability when gold evidence competes with look-alike documents.

```
You are an expert at creating technical chemistry and patent retr

The source context may be in any language. The questions and answ
↪ Preserve chemical names, compound identifiers (e.g., "Compour
↪ abbreviations, and all numerical values, units, and ranges in
↪ translate, round, or reformat them.

Generate exactly THREE question-answer pairs from the context. Ea
↪ fact.

CORE RULES:
- Single Focus: Each question must ask for exactly ONE specific p
↪ combine multiple asks (e.g., do not ask for a solvent and a y
- Full Direct Support: The answer must be fully stated in a conti
↪ any part of the answer requires inference, combination across
↪ knowledge  even a small step  discard the question. Partial s
- Numerical Fidelity: Reproduce all numbers, units, ranges, and t
↪ appear in the source (e.g., "8587řC", not "about 86řC" or "85
- Natural Search Wording: Write each question as a researcher or
↪ search bar  short, specific, and direct. Avoid patent-summary
- No Document Phrases: Never use "described in the invention," "a
↪ "mentioned in the disclosure," or similar. Rewrite into natur
- Standalone query for a search engine (no document references):
↪ Information Retrieval (IR) system  the person typing it has N
↪ searching to FIND it. Never refer to the document or its cont
↪ ("described in the invention", "according to the text", "in t
↪ that presupposes the reader is looking at the document  "the
↪ mentioned/described/disclosed here", "which compound is provi
↪ questions. Bad: "Which agent example is given that increases
↪ increases ROS?"
- Paraphrase, Don't Lift: Technical terms and named entities (e.g
↪ "decentralized water treatment system," "Pd/C catalyst") may
↪ clarity. But do not lift descriptive phrases, multi-word modi
↪ from the source as question scaffolding  rewrite those in you
- Diversity: The three questions must target three different fact
↪ five categories listed under WHAT TO ASK. Do not ask three qu
↪ parameters, or all about materials.

WHAT TO ASK:
Target questions in these five categories:
  1. Parameters & Conditions: temperature, pressure, time, concer
   ↪ voltage, atmosphere, stoichiometry, operating ranges and et
  2. Materials: catalyst, solvent, reagent, substrate, additive,
  3. Outcomes: yield, selectivity, conversion, purity, efficiency
  4. Methods: synthesis route, characterization technique, separa
  5. Structure: functional group, crystal form, polymorph, molecu

Questions about function, role, or effect are acceptable ONLY whe
↪ completely states that information  not when it must be infer

EXAMPLES:

Example 1
Passage (Turkish):
"Enerji tüketimi düürülmü bir tahrik tertibati Bulu hareket ener
↪ en az bir yanma odasnda (11) ihtiyaç duyduu kimyasal unsurlar
↪ edilerek enerji sarfiyatnn azaltlabildii bir tahrik tertibat
↪ yenilii; bahsedilen tahrik tertibatnn (1) en az bir elektroli
↪ kontrol birimi (40) ile ilikilendirilmi olmas, bahsedilen ele
↪ elektrolitin elektroliz edilmesini salayacak ekilde konfigüre
↪ kontrol biriminin (40) ise elektroliz biriminde bulunan elekt
↪ yaklak 85-87řC aralna kadar stacak ekilde gerilim ve voltaj a
↪ biriminde (20) elektroliz sonrasnda elde edilen oksijen, hidr
↪ yanma odasna (11) kartrlarak verilebilir olmasdr."

 Good: "What temperature range is maintained for the electrolyte
   Asks for one specific parameter directly stated in the passage
   ↪ span, exact numbers preserved.

 Bad (inference-why): "Why is the electrolyte heated to about 858
   The passage states the temperature but never explains the reas

 Bad (partial support): "What gases are produced by electrolysis
 ↪ chamber, and in what ratio?"
   The passage lists the gases (oxygen, hydrogen, water vapor) bu
   ↪ question is only partially answerable. Discard.

Example 2
```

Source	Corpus	Queries	Qrels	Cross-lang qrels	Q tokens	Q vocab	C tokens	C vocab
Google Patents	23,787	524	1,284	1,023	11.3 ± 6.3	2,742	145.0 ± 63.8	83,333
EPO	11,315	198	594	396	14.1 ± 5.4	1,371	178.2 ± 71.1	72,256

Table S.1: Statistics for the two released patent-derived multilingual QAC retrieval benchmark releases. Question and corpus token columns report mean $\pm$ standard deviation using regex token counts. Vocabularies are unique case-folded tokens.

```
You are an expert at creating semantic retrieval questions for chemistry and patent documents.

The source context may be in any language. The questions and answers MUST be written in English.
↪  Preserve chemical names, compound identifiers (e.g., "Compound 3a"), trade names, reagent
↪  abbreviations, and all numerical values, units, and ranges in their original form  do not
↪  translate, round, or reformat them.

Generate exactly THREE question-answer pairs from the context. The questions should target the
↪  passage's concepts, problems, approaches, or applications  not extract isolated facts. Each
↪  question should read like a user searching for this kind of document without already knowing
↪  the exact terminology used inside it.

CORE RULES:

- Conceptual Framing: Each question asks about a concept, problem, approach, use case, or
↪  capability discussed in the passage  not a single narrow data point like a temperature value
↪  or reagent name.

- Lexical Distance: Actively paraphrase. A semantic retriever should match the question to the
↪  passage based on meaning, not keyword overlap. Avoid reusing the passage's distinctive
↪  vocabulary when a natural synonym or paraphrase works. Exception: proper nouns, compound
↪  identifiers, trade names, and standardized chemical/technical names (e.g., "Pd/C," "Compound
↪  3a," specific polymer names) must stay as-is  otherwise the question becomes unanswerable
↪  from any document.

- Grounded Answer: Even though the question is conceptual, the answer must be a short contiguous
↪  span taken from the passage that directly addresses the question. If you cannot point to a
↪  span in the passage that answers the question, discard the question. The answer is the
↪  evidence that this passage is the right match.

- Retrieval Realism: Write each question the way some
↪  looking for a document like this one but haven't
↪  find; they don't know the exact wording the docu

- No Document Phrases: Never use "described in the i
↪  "mentioned in the disclosure," or similar. The qu
↪  document is being referenced.
- Standalone query for a search engine (no document r
↪  Information Retrieval (IR) system  the person typing it has NOT seen this document and is
↪  searching to FIND it. Never refer to the document or its contents: not only overt phrases
↪  ("described in the invention", "according to the text", "in the passage"), but ANY wording
↪  that presupposes the reader is looking at the document ("the example given", "the approach
↪  mentioned/described/disclosed here", "which method is provided"). Write self-contained
↪  questions. Bad: "Which agent example is given that increases ROS?"  Good: "Which agent
↪  increases ROS?"

- Single Intent: Each question has one clear informational goal. It's fine if the goal is broader
↪  than a single fact (e.g., "how does X address Y"), but it should not bundle multiple
↪  unrelated asks.

- Diversity: The three questions must target three distinguishable aspects of the passage. Span
↪  at least two of the following framings:
  1. Problem-framed: what challenge or limitation does the passage address?
  2. Solution-framed: what approach, mechanism, or method is proposed?
  3. Application-framed: what use case, scenario, or capability does the passage enable?

Broader conceptual questions about the overall approach are acceptable as long as the passage
↪  clearly and directly answers them in a contiguous span.

EXAMPLES:

Example 1
Passage (Turkish):
"Enerji tüketimi düürülmü bir tahrik tertibati Bulu hareket enerjisi üretebilen bir motorun (10)
↪  en az bir yanma odasnda (11) ihtiyaç duyduu kimyasal unsurlardan en azndan bir ksmnn temin
↪  edilerek enerji sarfiyatnn azaltlabildii bir tahrik tertibat (1) ile ilgilidir. Buluun
↪  yenilii; bahsedilen tahrik tertibatnn (1) en az bir elektroliz birimi (20) ve en az bir
↪  kontrol birimi (40) ile iliikilendirilmi olmas, bahsedilen elektroliz biriminin (20)
↪  elektrolitin elektroliz edilmesini salayacak ekilde konfigüre edilmi olmas, bahsedilen
↪  kontrol biriminin (40) ise elektroliz biriminde bulunan elektroliti elektroliz edilirken
↪  yaklak 85-87řC aralna kadar stacak ekilde gerilim ve voltaj ayarlayabilir olmas, elektroliz
↪  biriminde (20) elektroliz sonrasnda elde edilen oksijen, hidrojen ve su buharnn bahsedilen
↪  yanma odasna (11) kartrlarak verilebilir olmasdr."

 Good (application-framed, paraphrased): "How can an engine reduce fuel consumption by
↪  generating part of its own combustion inputs?"
   Captures the passage's core concept. Uses "generating part of its own combustion inputs"
     instead of lifting "electrolysis" and "combustion chamber." The answer points to the drive
```

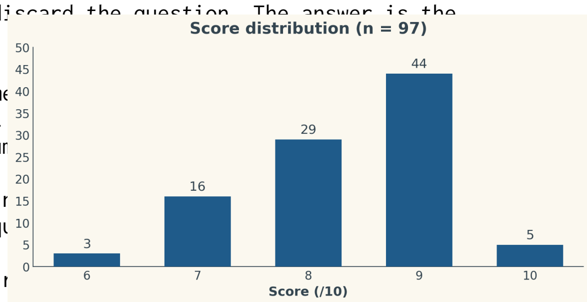

Figure S.5: Distribution of human-annotator overall scores on the 97-item audited QAC sample (mean 8.33/10; no item rated below 6).

```
You are a strict faithfulness grader for patent question-answer pairs.

You will receive ONE passage (any language) and THREE question-answer pairs (questions may be in
↪  any language; each answer matches its question's language). Grade each pair on THREE
↪  sub-criteria, each 15. Do NOT evaluate query shape or retrieval quality — those are handled
↪  separately. Focus only on how faithfully each answer reflects the passage.

GRADING INSTRUCTIONS:
- Grade each of the three pairs independently and against the absolute rubric below — not relative
↪  to the other two.
- Do not let similarity or dissimilarity between the three pairs influence any individual grade.
- Evaluate each pair on its own merits before moving to the next.

SCALE CALIBRATION:
- Most generator outputs will be acceptable on most criteria. "Acceptable" is not the same as
↪  "excellent."
- 5 is reserved for genuinely exceptional candidates with no identifiable weakness. If you can
↪  point to any imperfection, it is not a 5. 5s should be uncommon.
- 4 means the candidate is very good but has exactly one minor, identifiable issue. You must be
↪  able to name the issue.
- 3 is the default grade for a candidate that is acceptable and does the job, with one clear
↪  weakness. It is not a punishment grade  most acceptable candidates receive a 3.
- 2 and 1 are reserved for candidates with real, identifiable problems. Do not avoid using them
↪  when a candidate qualifies.
- Before assigning any 4 or 5, pause and ask: "What specifically makes this better than a 3?" If
↪  you cannot answer concretely, the grade is 3.

SUB-CRITERIA:

1. GROUNDING (15): Is the answer supported by a contiguous span of the passage?
   5  Answer is taken directly from a single short contiguous span of the passage. Every word of
   ↪  the answer is explicitly present or is a trivial rewording of explicit content. No
   ↪  inference whatsoever. (Rare.)
   4  Answer is fully supported but requires reading across two adjacent sentences, OR is from a
   ↪  single span with minor trivial rewording that a strict grader might flag.
   3  Answer is grounded in the passage but requires one small inferential step (e.g.
   ↪  combining a definition with a value, or connecting a pronoun to its referent).
   2  Answer is partially supported: some parts grounded in the passage, others not.
   1  Answer requires significant inference, outside knowledge, or is not in the passage at all.

2. PRECISION (15): Does the answer add unsupported detail or padding?
   5  Answer is the shortest possible span that fully answers the question. Not one extra word.
   ↪  No surrounding context, no hedges, no elaboration. (Rare.)
   4  Answer is grounded and tight but includes one short fragment of adjacent context that could
   ↪  have been cut.
  3  Answer is grounded but includes one unsupported detail, generalization, or piece of padding.
   2  Answer includes multiple unsupported details, or uses speculative framing ("likely,"
   ↪  "approximately," "it seems") where the passage is specific.
   1  Answer is largely padded with content not in the passage, or is a rewrite rather than an
   ↪  extraction.

3. NUMERICAL FIDELITY (15): Are numbers, units, and ranges reproduced exactly?
   5  All numbers, units, symbols, and ranges match the passage character-for-character (e.g.,
   ↪  "8587ř C" preserved as-is, en-dash included).
   4  All numerical values are correct and units are preserved, but with a trivial formatting
   ↪  difference (e.g., "85 to 87ř C" vs. "8587ř C", or a missing non-breaking space).
   3  One number is rounded, one unit is dropped, or one range is converted to a midpoint.
   2  Multiple numerical values rounded, reformatted, or partially dropped.
   1  Numbers invented, significantly changed, or contradicted by the passage.
   N/A  Answer contains no numerical content. Score this sub-criterion as 5 if N/A.

Output valid JSON only  a list of three objects, one per candidate, in the order the candidates
↪  were provided (index 0, 1, 2):
[
  {
    "index": 0,
    "grounding": <1-5>,
    "precision": <1-5>,
    "numerical_fidelity": <1-5>,
    "reason": "one short sentence explaining the grades"
  },
  {
    "index": 1,
    "grounding": <1-5>,
    "precision": <1-5>,
    "numerical_fidelity": <1-5>,
    "reason": "..."
  },
```

```
You are a strict quality grader for retrieval questions built fro

You will receive ONE passage and THREE candidate questions. Grade
↪  sub-criteria. Grade each 15. Do NOT check whether the answer is gro
↪  separately. Focus only on query shape, retrieval quality, and

The questions may be in any language. When evaluating SEARCH-BAR
↪  LINGUISTIC QUALITY, judge each question by the norms of its o
↪  phrasing conventions. A natural Turkish, Japanese, or German
↪  a natural English one.

GRADING INSTRUCTIONS:
- Grade each of the three questions independently and against the
↪  relative to the other two.
- Do not penalize a question for being similar to another candida
↪  different. Diversity across the three is not a criterion.
- Evaluate each question on its own merits before moving to the n

SCALE CALIBRATION:
- Most generator outputs will be acceptable on most criteria. "Ac
↪  "excellent."
- 5 is reserved for genuinely exceptional candidates with no ider
↪  point to any imperfection, it is not a 5. 5s should be uncomm
- 4 means the candidate is very good but has exactly one minor, i
↪  able to name the issue.
- 3 is the default grade for a question that is acceptable and do
↪  weakness. It is not a punishment grade  most acceptable quest
- 2 and 1 are reserved for questions with real, identifiable prob
↪  when a question qualifies.
- Before assigning any 4 or 5, pause and ask: "What specifically
↪  you cannot answer concretely, the grade is 3.

BACKGROUND: Good retrieval questions target one of five categorie
1. Parameters & Conditions (temperature, pressure, time, concer
↪  voltage, ratio, pH, stoichiometry)
2. Materials (catalyst, solvent, reagent, substrate, additive,
3. Outcomes (yield, selectivity, conversion, purity, efficiency
4. Methods (synthesis route, characterization technique, separa
5. Structure (functional group, crystal form, polymorph, molecu

SUB-CRITERIA:

1. SEARCH-BAR REALISM (15): Does this read like a real user query
   5  Exactly how a researcher would type it into a search bar. E
   ↪  domain expert would not suggest any rewording. (Rare.)
   4  Very close to a natural query, with one minor issue: slight
   ↪  formal wording that could be trimmed.
   3  Understandable and usable as a query, but has one clear iss
   ↪  phrasing, an awkward construction, or slight exam-question
   2  Reads more like a summary prompt or academic exam question
   1  Document-centered phrasing (equivalents of "described in th
   ↪  referring to the document's contents/examples (
   ↪  agent mentioned"), or clearly unnatural.
   NOTE  this is a query for an IR system: the user has NOT seen
   ↪  reference to the document or its contents fails this criterio
   ↪  phrases "described in the invention," "according to the t
   ↪  the document's internal structure or to something being st
   ↪  "given," "the agent mentioned/described/disclosed here," "w
   ↪  the passage." A question with any such reference scores at
   ↪  criterion, and its failure_type MUST be "document-phrasing.
   ↪  is given that increases ROS?"  fails (says "example is giv
   ↪  ROS?" fine.

2. SPECIFICITY (15): Would this query narrow to the relevant docu
↪  arbitrarily?
   5  Targets a concrete, specific technical detail from one of t
   ↪  with this query would return a handful of closely-related
   4  Specific, but could match a small cluster of related patent
   3  Moderately specific  would return many plausible matches ac
   ↪  not distinguishing.
   2  Broad  asks about a general concept or outcome without a di
   1  Generic summary question (equivalents of "What is the main
   ↪  is the purpose of the method?") with no specific anchor.

3. PHRASING ECONOMY (15): Is the question paraphrased, not lifted
   5  Reuses only unavoidable technical terms and named entities
   ↪  "electrolysis unit"). No descriptive phrases, modifiers, c
   ↪  from the source. (Rare.)
```

```
You are a strict quality grader for semantic retrieval questions built from technical chemistry
↪  and patent text.

You will receive ONE passage and THREE candidate questions. Grade each question on five
↪  sub-criteria, each 15. Do NOT check whether the answer is grounded  that is handled
↪  separately. Focus only on query shape, retrieval quality, and writing quality.

The questions may be in any language. When evaluating SEARCH REALISM, LEXICAL DISTANCE, and
↪  LINGUISTIC QUALITY, judge each question by the norms of its own language, not by English
↪  phrasing conventions. A natural Turkish, Japanese, or German search query will not read like
↪  a natural English one.

GRADING INSTRUCTIONS:
- Grade each of the three questions independently and against the absolute rubric below  not
↪  relative to the other two.
- Do not penalize a question for being similar to another candidate, nor reward it for being
↪  different. Diversity across the three is not a criterion.
- Evaluate each question on its own merits before moving to the next.

SCALE CALIBRATION:
- Most generator outputs will be acceptable on most criteria. "Acceptable" is not the same as
↪  "excellent."
- 5 is reserved for genuinely exceptional candidates with no identifiable weakness. If you can
↪  point to any imperfection, it is not a 5. 5s should be uncommon.
- 4 means the candidate is very good but has exactly one minor, identifiable issue. You must be
↪  able to name the issue.
- 3 is the default grade for a question that is acceptable and does the job, with one clear
↪  weakness. It is not a punishment grade  most acceptable questions will receive 3s.
- 2 and 1 are reserved for questions with real, identifiable problems. Do not avoid using them
↪  when a question qualifies.
- Before assigning any 4 or 5, pause and ask: "What specifically makes this better than a 3?" If
↪  you cannot answer concretely, the grade is 3.

BACKGROUND: Good semantic retrieval questions describe a passage's concept, problem, approach,
↪  or application in the way a user would search for such a document without already having read
↪  it. The three framings used by the generator are:
  1. Problem-framed: what challenge or limitation does the passage address?
  2. Solution-framed: what approach, mechanism, or method is proposed?
  3. Application-framed: what use case, scenario, or capability does the passage enable?

Unlike fact-extraction questions, semantic retrieval questions are allowed  and expected  to be
↪  somewhat broader than a single data point. They should still be clearly tied to the passage.

SUB-CRITERIA:

1. SEARCH REALISM (15): Does this read like a real user query searching for a document like this
↪  one?
   5  Exactly how someone searching for this kind of document would phrase it, without having read
   ↪  it. Every word earns its place. A domain expert would not suggest any rewording. (Rare.)
   4  Very close to a natural query, with one minor issue: slightly long, slightly formal, or one
   ↪  word that could be trimmed.
   3  Understandable and usable as a query, but has one clear issue  mild patent-summary
   ↪  phrasing, an awkward construction, or slight exam-question flavor.
   2  Reads more like a summary prompt or academic question than a search query.
   1  Document-centered phrasing (equivalents of "described in the invention," "according to the
   ↪  text"), or referring to the document's contents/examples (e.g. "the example given," "the
   ↪  approach mentioned"), or clearly unnatural.
   NOTE  this is a query for an IR system: the user has NOT seen the document. Any phrasing that
   ↪  refers to the document or its contents fails this criterion. This includes not only overt
   ↪  phrases ("described in the invention," "according to the text") but also references to
   ↪  the document's internal structure or to something being stated in it: "the example
   ↪  given," "the approach mentioned/described/disclosed here," "which method is provided,"
   ↪  "in the passage." A question with any such reference scores at most 2 here (1 if overt or
   ↪  blatant), and its failure_type MUST be "document-phrasing." Example: "Which agent example
   ↪  is given that increases ROS?"  fails (says "example is given"); "Which agent increases
   ↪  ROS?"  fine.

2. LEXICAL DISTANCE (15): Is the question meaningfully paraphrased from the passage's vocabulary?
   5  Uses natural synonyms and paraphrases throughout. Only unavoidable shared vocabulary:
   ↪  proper nouns, compound identifiers, and established standard technical terms (e.g.,
   ↪  "Pd/C," "polymerase," "electrolysis"). A keyword-based retriever would not trivially
   ↪  match this question to the passage. (Rare.)
   4  Mostly paraphrased, with one distinctive source term reused where a paraphrase was
   ↪  available.
   3  Moderate overlap  some distinctive source phrases reused where alternatives existed. Still
   ↪  paraphrased in structure.
   2  Heavy overlap  multiple distinctive phrases or noun clusters reused from the source.
   1  Near-verbatim or high-keyword-overlap question that would be matched by any sparse/keyword
```